

Multiple Integration

Problem Sheet 4

1. Consider the repeated integral

$$I = \int_0^2 \int_0^1 (x + y) \, dx \, dy.$$

Sketch the region of integration and evaluate I .

2. Evaluate the following repeated integrals.

(a) $\int_0^1 \int_0^3 z^2 \, dx \, dz,$

(b) $\int_0^4 \int_0^2 x \, dy \, dz,$

(c) $\int_1^2 \int_2^3 (x^2 y + 1) \, dx \, dy,$

(d) $\int_{-1}^1 \int_2^3 \frac{y}{x} \, dx \, dy.$

3. For each of the following integrals, sketch the region of integration and write an equivalent repeated integral with the order of integration reversed. **You do not need to evaluate the resulting integral.**

(a) $\int_0^1 \int_2^{4-2x} dy \, dx.$

(b) $\int_0^1 \int_y^{\sqrt{y}} dx \, dy.$

(c) $\int_0^{\ln 2} \int_{e^y}^2 dx \, dy.$

(d) $\int_0^{3/2} \int_0^{9-4x^2} 16x \, dy \, dx.$

(e) $\int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} 3y \, dx \, dy.$

(f) $\int_1^e \int_0^{\ln x} xy \, dy \, dx.$

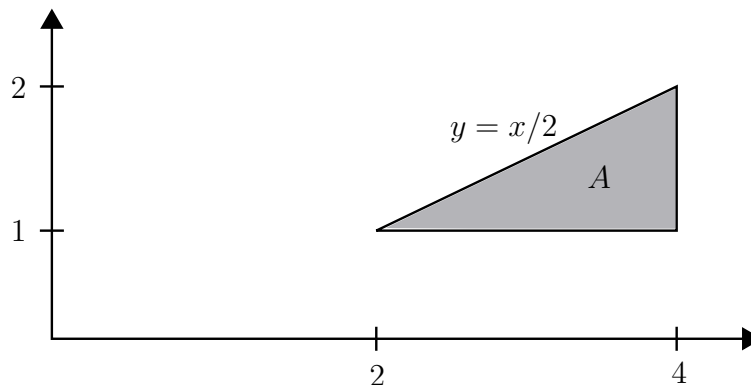
(g) $\int_0^{\pi/6} \int_{\sin x}^{1/2} xy^2 \, dy \, dx.$

(h) $\int_0^{\sqrt{3}} \int_0^{\tan^{-1} y} \sqrt{xy} \, dx \, dy.$

4. Consider the multiple integral

$$I = \iint_A F(x, y) \, dA,$$

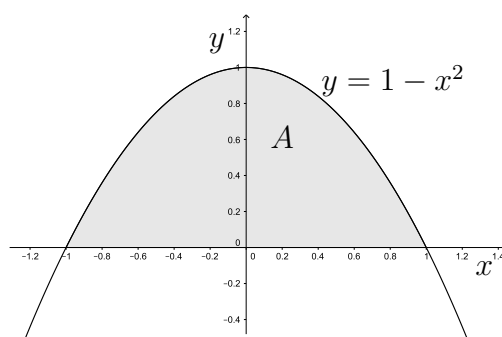
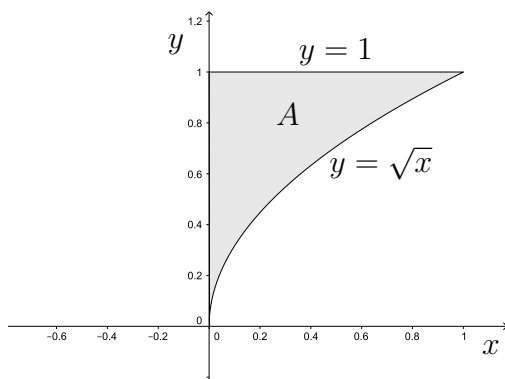
where $F(x, y)$ is an arbitrary function of x and y . Express I as a repeated integral for the region A shown below.



State the repeated integral and the limits of integration when

- (a) the inner integral is with respect to x ;
- (b) the inner integral is with respect to y .

5. Repeat the previous question for the regions shown in the figures below.



6. Repeat the previous question for the regions described below. It may help to sketch the region.

- (a) The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 9$.
- (b) The region bounded by $y = \tan x$, $x = 0$ and $y = 1$.
- (c) The region bounded by $y = e^{-x}$, $y = 1$, and $x = \ln 3$.
- (d) The region bounded by $y = 0$, $x = 0$, $y = 1$, and $y = \ln x$.
- (e) The region bounded by $y = 3 - 2x$, $y = x$, and $x = 0$.

7. Sketch the region and evaluate the following integrals:

(a) $\int_0^\pi \int_0^x x \sin y \, dy \, dx.$

$$(b) \int_0^\pi \int_0^{\sin x} y \, dy \, dx.$$

$$(c) \int_1^{\ln 8} \int_0^{\ln y} e^{x+y} \, dx \, dy.$$

$$(d) \int_1^2 \int_y^{y^2} dx \, dy.$$

$$(e) \int_0^1 \int_0^{y^2} 3y^3 e^{xy} \, dx \, dy.$$

$$(f) \int_1^4 \int_0^{\sqrt{x}} \frac{3}{2} e^{y/\sqrt{x}} \, dy \, dx.$$

8. Let A denote the region in the xy -plane enclosed by the lines $x = 0$, $y = 1$, $x = y^2$. Sketch the region A and evaluate the integral

$$\iint_A (2xy - x^2) \, dA$$

by expressing it as a repeated integral

- (a) where the inner integral is with respect to x ;
- (b) where the inner integral is with respect to y .

Confirm that the value of the integral is $5/42$.

9. Sketch the region of integration of the repeated integral

$$I = \int_0^3 \int_1^{\sqrt{4-y}} (x+y) \, dx \, dy,$$

and confirm that I has the value $241/60$.

Check this answer by first changing the order of integration and then evaluating the resulting repeated integral.

10. Consider the integral I below.

$$I = \iint_A (5x^2 + 2y^2) \, dA,$$

where A is the triangular region bounded between $A(1, 1)$, $B(2, 0)$ and $C(2, 2)$. Sketch the region A and show that the value of the integral is $33/2$.

11. For each of the following integrals, sketch the region of integration, reverse the order of integration, then evaluate the integral.

$$(a) \int_0^\pi \int_x^\pi \frac{\sin y}{y} \, dy \, dx.$$

$$(b) \int_0^1 \int_y^1 x^2 e^{xy} \, dx \, dy.$$

$$(c) \int_0^3 \int_{\sqrt{x/3}}^1 e^{y^3} \, dy \, dx.$$

$$(d) \int_0^{1/16} \int_{y^{1/4}}^{1/2} \cos(16\pi x^5) \, dx \, dy.$$

12. Evaluate

$$I = \iint_A (x + 2y)^{-1/2} \, dA,$$

where A is the region given by $x - 2y \leq 1$ and $x \geq y^2 + 1$.

Prerequisites:

$P(n) = \{\text{questions with prerequisite Parts } \leq n\}$

$A = \text{all of PS4.}$

$P(11) = \{1, 2\}$

$P(12) = A$